

First Semester 2026-27
Course Handout Part II

Date: 5 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : MPBA G507
Course Title : Programming for Analytics
Instructor-in-charge : [Prof. Pabitra Biswas](#)
Co-Instructor : [Prof. Debasmita Dey](#)

1. Course description:

Variables & data types, operators, conditional statements, iteration statements, functions, Objects, and classes for Python; Data structures – vectors, matrices, lists, tuples, dictionaries, data frames, Packages: using existing packages and creating new packages, Data loading: from CSV, MS Excel, JSON & SQL databases Data wrangling: describing, summarizing, manipulating, and handling missing values of data in data frames; perform an operation such as delete values, find unique values, drop duplicate values, grouping, and performing loop operations over columns or rows; merging data frames; handling categorical data; handling time-series data, handling text data, web scraping, and handling images. Basics of version control using Github. Primary languages are R and Python.

2. Scope and Objective:

1. To develop proficiency in programming to automate analytical workflows, including data cleaning, transformation, and report generation, reducing manual effort and increasing accuracy.
2. To gain hands-on experience with programming tools and libraries (e.g., Python's pandas, R's tidyverse) to efficiently process, manipulate, and analyze large and complex datasets beyond the capabilities of traditional tools.
3. To apply programming best practices to ensure reproducibility, and transparency, for analytical processes, enabling consistent validation and collaboration across teams.

3. Course learning outcomes:

CLO1. To be able to understand different types of data structures and implement them using R and Python.

CLO2. To be able to load datasets using R and Python language.

CLO3. To be able to transform and manipulate datasets using R and Python language.

4. Text books and references:

TB1. Wickham, H., Centinkaya-Rundel, M., & Grolemund, G. (2023). R for Data Science (2e)
<https://r4ds.hadley.nz/>

TB2. Peng, Robert (2016). R Programming for Data Science
<https://www.cs.upc.edu/~robert/teaching/estadistica/rprogramming.pdf>.

TB3. Downey, Allen, Jeffrey Elkner, and Chris Meyers. How to Think Like a Computer Scientist: Learning with Python (ISBN: 9780971677500)
<http://www.greenteapress.com/thinkpython/thinkCSpy/thinkCSpy.pdf>.

RB1. Lander, J. P. (2018). *R for Everyone: Advanced Analytics and Graphics*, 2/E. Pearson Education.

RB2. JD Long & Paul Teetor (2019). *R Cookbook: Proven Recipes for Data Analysis, Statistics, and Graphics* Second Edition

RB3. John Whittington (2023). *Python from the Very Beginning* (ISBN-13: 978-0-9576711-5-7, 238 Pages), Second Edition

5. Course Format & Weekly Workflow

Note: Carrying a laptop is mandatory for every class.

R Programming Module

Wk no.	Module no.	Description	Ref.	Learning Outcome
1	Introduction to R programming and Data structures, variables, and data types	• Installing R and RStudio • Arithmetic and Logical Operations • Using Functions • Creating Variables • Numeric, Character and Logical Data • Vectors, Data Frames, Factors, Sorting Numeric, Character, and Factor Vectors	TB1: Ch 2,3,8,12,16. TB2: Ch - R Nuts and Bolts, Factors, Debugging, Functions, Vectorized Operations	Understand the working environment of R studio console and fundamental syntax of R through readings, practice exercises, demonstrations, and writing R code
2	R packages and scripts	• Installing and loading packages • Setting up your working directory • Downloading and importing data • Working with missing data • Writing R scripts • Adding comments and	TB1: Ch- 1,3,6,7,18 TB2: Ch- Managing Data Frames with the dplyr package, Getting Started with R, Getting	Import a variety of data formats into R using RStudio

		documentation	Data In and Out of R, Using the readr Package, Interfaces to the Outside World, R Nuts and Bolts, Subsetting R Objects, Coding Standards for R	
3	Descriptive statistics in R and Statistical graphs	• Measures of central Tendency, variability, Skewness and kurtosis, Summary functions, describe functions, and descriptive statistics by group • Correlations • Scatter Plots, Box Plots • Histograms	TB1: Ch 1,3,9,10,11,13,25 TB2: Data Analysis Case Study: Changes in Fine Particle Air Pollution in the U.S., Simulation, Managing Data Frames with the dplyr package	Analyze a data set statistically in R and present findings using the appropriate R packages in various formats (tables and figures)
4	Working with messy data	• Messy Data • Renaming Columns • Attaching / Detaching • Tabulating Data: Constructing Simple Frequency Tables	TB1: Ch - 3,5,13,16 TB2: Ch - R Nuts and Bolts, Loop Functions, Managing Data Frames with the dplyr package	Prepare or manipulate complex/messy data for analysis ready structure for data processing pipeline.
5	Iteration and Conditional statements	• while loops • for loops • If / else • Boolean logical operators	TB1: Ch- 12, 27 TB2: Ch - Vectorized Operations, Control	Apply critical programming language concepts such as iteration, control structures, functions, and

			Structures, for Loops, Nested for loops, while loops	Boolean operators by writing R programs and through practice examples
6	Data querying: SQL and R	• Writing SQL statements in R • Using the Select, From, Where, Is, Like, Order By, Limit, Max, Min SQL functions	TB1: Ch – 21 TB2: Ch - <i>Managing Data Frames with the dplyr package</i>	Query data using SQL and R
7	Interactive reporting with Rmarkdown	• RMarkdown basics • Text formatting • Code chunks • YAML header • Preview of notebooks, presentations, websites, and dashboards	TB1: Ch – 28,29	Present the analysis findings into high-quality documents, reports, presentations and dashboards for business stakeholders
MID-SEMESTER EXAMINATIONS				

Python Programming Module

Part I. Essentials of Python

S. No.	Topic	Topic Description
1.	Beginning of Python Programming Journey	• What is programming? • What is Python? • Why is Python relevant to IT? • Other Languages • Downloading & Installing Jupyter Notebook • Use JupyterLab and Jupyter Notebooks • Use Colab
2.	Basic Syntax in Python	• Syntax • Data Types • Expressions, numbers, and type conversions • Defining Functions • Branching with if Statements

3.	Loops & Strings	• What is a while loop? • What is an Infinite Loop and How to break it? • Where and How to use a for Loop • What is a string? • What is a list? • Lists and Tuples • What is a dictionary? • Lambda with max and min
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Part II. Working with Data in Python

S. No.	Topic	Topic Description
4	Importing and Reading a file using the CSV module	• Import data from csv file • Reading data from csv file • Basic Data Analysis • Understand errors • What is a module in Python? • Accessing modules
5	The pandas Library	• Installing Packages with pip in Python • Saving Requirements File in Python • Creating Pandas DataFrames in Python • Investigating Data in a Pandas DataFrame • Selecting Data in a Pandas DataFrame • Manipulating Pandas DataFrames • Updating Pandas DataFrame Data • Applying Functions in a Pandas DataFrame • Spark and PySpark DataFrames in Python
6	Git and Git Concepts	• What is Version Control • Version Control with GitHub
7	The numpy Library	• Creating array Array Indexing & Slicing Data Types • Copy vs View Array reshape Array Join & Split Array Search & Sort Array Filter



Part III. Visualization with Python

S. No.	Topic	Topic Description
8	The Matplotlib Library	Types of plots Line Graph Bar Chart Histogram Scatter plot Box Plot Violin Plot Subplots Heatmaps Saving and exporting plots
9	The Seaborne Library	Using Seaborn with Matplotlib Customizing Seaborn Plots Changing Figure Aesthetic Removal of Spines Changing the figure Size Scaling the plots Setting the Style Temporarily Color Palette Diverging Color Palette Sequential Color Palette Multiple plots with Seaborn Relational Plots Categorical Plots Distribution Plots

6. Evaluation Scheme:

R Module

Component	Weightage	Duration	Marks	Date and time	Comments
In-class/lab Assignment <i>*Data will be provided</i>	10%	In lecture/lab session	20	Weekly once	Open book/notes/ Internet
Group Project <i>*Dataset must be decided by student groups and submitted latest by 21st August EOD</i>	10%	Till MID Sem	10	Submission: 28 th Sept Presentation : 29 th Sept and 1 st Oct	Open book/notes/ Internet
Pre-MID Sem Quiz	10%	30 mins	10	3 rd Week of September	Closed book/notes/ Internet
Mid Semester	20%	90 min	60	2 nd Week of	Closed book/notes/

				October	Internet
Total			100		

Python Module

Component	Duration	Marks	Weightage (%)	Date & Time	Remarks
Comprehensive exam	180 min.	60	45%	TBD	*
Quiz (Post-Mid)	15 min	10	15%	2 nd week of Nov 26	*
In-Class Assignments & Group Project	3 Months (During Class/Lab)	15 +15	20% +20%	TBD	*
Total		100	100%**		

** : 50% weightage would be considered during final grading.

Final Grading: The complete course will be worth 200 marks. R Module evaluation will be of 100 marks (50% weightage) and Python module will be of 100 marks (50% weightage)

7. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components. The “In-class/lab Assignment” from pre-MID assessment component will be conducted either on lecture session and/or in lab session. The assignment date will be announced at least one day prior and an absence in the same will lead to zero marks for that assignment. These assignments are linked with your attendance in class. No makeup assignments/quizzes are allowed. Please refer to the Makeup Policy section for more information.
8. **Chamber consultation hours:**
 - **Prof. Debasmita Dey**- Mon & Wed: 11AM – 12:30 PM/ pre-decided appointment timings
 - **Prof. Pabitra Biswas** - Appointments via email / face-to-face discussions
9. **Makeup Policy:** Makeup is applicable strictly for Mid-Sem or comprehensive exam only for cases related **Emergency** medical situations in campus. The medical documents (prescription/certificates etc.) prepared/attested by BITS Medical Centre, clearly specifying “*the student is unfit to attend the test*”, will be essential and non-negotiable. The final decision will be at the discretion of the IC.

Instructor-in-charge

MPBA G507